

PROGRAM:

```
#include <stdio.h>

int main() {

    int n, i, search, f = 0, low, high, mid, a[20];

    printf("\n***Binary Search***\n");

    printf("\nEnter the number of elements: ");

    scanf("%d", &n);

    // Input array elements in ascending order

    printf("Enter %d numbers in ascending order:\n", n);

    for (i = 0; i < n; i++) {

        printf("a[%d] = ", i);

        scanf("%d", &a[i]);

    }

    printf("Enter the search element: ");

    scanf("%d", &search);

    low = 0;

    high = n - 1;

    // Binary search

    while (low <= high) {

        mid = (low + high) / 2;

        if (search < a[mid])

            high = mid - 1;

        else if (search > a[mid])

            low = mid + 1;

        else {

            f = 1;

            printf("Element %d found at index %d.\n", search, mid);

            break;

        }

    }

}
```

```
    if (f == 0)

        printf("Element %d is not present in the array.\n", search);

    return 0;

}
```

OUTPUT:

Binary Search

Enter the number of elements: 4

Enter 4 numbers in ascending order:

a[0] = 23

a[1] = 45

a[2] = 67

a[3] = 87

Enter the search element: 45

Element 45 found at index 1.

Binary Search

Enter the number of elements: 4

Enter 4 numbers in ascending order:

a[0] = 56

a[1] = 78

a[2] = 87

a[3] = 90

Enter the search element: 10

Element 10 is not present in the array.